06B
THERMOCHEMISTRY

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 6

1. Define and apply energy, work, and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state functions and path functions.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Summarize the roles of enthalpy and heat flow in chemical reactions.
6. Calculate pressure-volume work.
7. Calculate the heat required to change the temperature of a substance.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of different reactions.
10. Apply standard enthalpies of formation and Hess's law to calculate the enthalpy of a reaction.

Energy and Enthalpy

6.4

constant atmospheric pressure

Most chemical experiments and reactions are best described as *open systems*.

- Exchange of *both* heat (q) and work (w) between the system and surroundings.
- Chemical reactions are often carried out on the "benchtop" where they are exposed to a constant (atmospheric) pressure, so

$$w = -P \cdot \Delta V \neq \text{zero}$$

- These conditions give rise to a very *useful and measurable* property called *enthalpy*.

Enthalpy (H) is a state function relating internal energy (ΔU), pressure (P), and volume (V).

From first law: $\Delta U = q + w$; if P constant: $q \rightarrow q_p$, $w = -P \cdot \Delta V$

$\Delta U = q_p - P \cdot \Delta V$

$q_p = \underbrace{\Delta U + P \cdot \Delta V}$; define right side as $\Delta H \equiv \Delta U + P \cdot \Delta V$

$q_p = \Delta H$

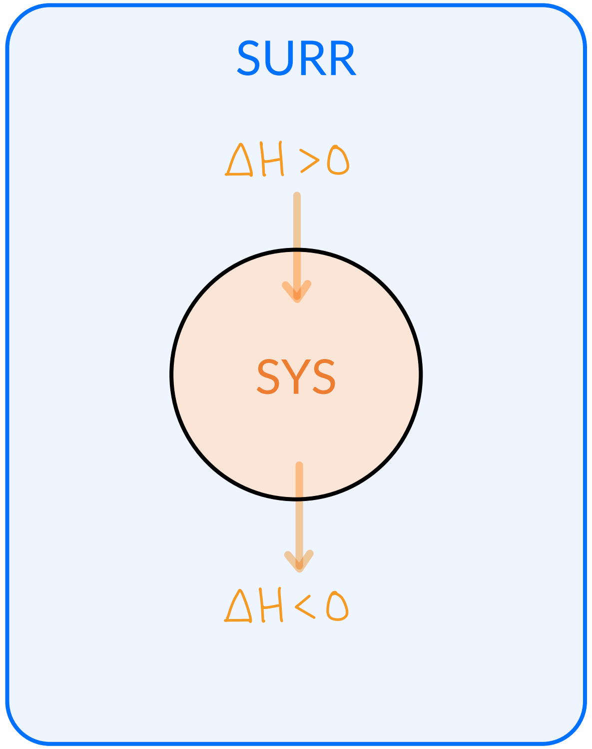
Energy and Enthalpy

6.4

The change in enthalpy for a reaction (ΔH_{rxn}) is *defined* as the flow of heat (q) when the chemical reaction is carried at a constant (atmospheric) pressure.

$q_{p,rxn} \leftrightarrow \Delta H_{rxn}$ synonymous at constant pressure!

Process	System/Rxn	Surroundings	Sign of ΔH_{rxn}
Exothermic	*Releases* heat $T_{sys} \downarrow$	*Absorbs* heat $T_{surr} \uparrow$	$\Delta H_{\text{rxn}} < 0$ ⊖
Endothermic	*Absorbs* heat $T_{sys} \uparrow$	*Releases* heat $T_{surr} \downarrow$	$\Delta H_{\text{rxn}} > 0$ ⊕



Energy and Enthalpy

Chemistry in Context

Recall that heat (q) is the transfer of energy *due to a temperature difference* between two objects. Heat will always flow until the two objects reach the same temperature, or a state called thermal equilibrium.

You should recognize that heat is an incredibly easy quantity for us to measure in the lab because it is related to changes in temperature. All we need is a thermometer!

The next several sections are all related to *different ways* to measure or calculate ΔH_{rxn}. There are three general methods:

1) Experimentally through calorimetry (measuring temperature changes)
2) Hess's Law: taking advantage of state function properties
3) Theoretically through standard enthalpies of formations, $\Delta H_{\mathrm{f}}^{\mathrm{o}}$

Specific Heat

6.5

In the *absence* of a phase change, the amount of heat (q) required to change the temperature (ΔT) of some quantity (m) of substance is given by

$$q = m \cdot c \cdot \Delta T$$

heat in J or kJ

mass in grams, g

temperature in °C or K

specific heat in $\frac{J}{g \cdot °C} = \frac{J}{g \cdot K}$

Specific heat capacity: (*also called specific heat,* c)

- The amount of energy required to increase the temperature of *exactly* 1 g of a substance by *exactly* 1 °C = 1 K.
- Also comes in a per mole variety called the *molar heat capacity*.

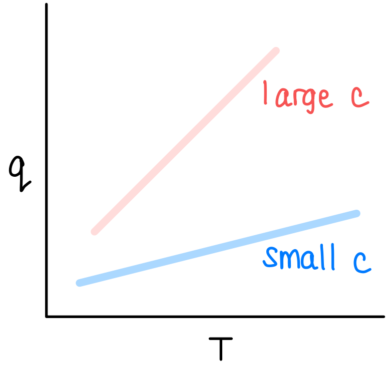
$$\frac{J}{g \cdot °C} \xrightarrow[\text{conversion}]{\text{molar mass}} \frac{J}{mol \cdot °C}$$

Specific Heat

6.5

note: table provided when needed!

Notes:
1) Liquid water has a *large* value of c.
2) Many metals tend to have *small* values of c.
3) We assume c does not change with temperature (not technically true).



Covalent Substance	Specific Heat $\left(\frac{J}{g \cdot °C}\right)$
CO_2 (*g*)	0.852
CO (*g*)	1.04
H_2 (*g*)	14.4
N_2 (*g*)	1.04
O_2 (*g*)	0.922
H_2O (*g*)	2.042
H_2O (*ℓ*)	4.184
H_2O (*s*)	2.089

TABLE 6.3 Specific Heat Capacities of Selected Substances

Metal Substance	Specific Heat $\left(\frac{J}{g \cdot °C}\right)$
Au (*s*)	0.129
Pb (*s*)	0.13
Sn (*s*)	0.22
Ag (*s*)	0.24
Na (*s*)	0.293
Cu (*s*)	0.385
Zn (*s*)	0.388
Fe (*s*)	0.442
Cr (*s*)	0.45
Co (*s*)	0.46
Al (*s*)	0.892
Mg (*s*)	1.0

EXAMPLE PROBLEM

Learning Objective(s): 6.7

What is the specific heat capacity (c) of an unknown metal if it takes 1155 J of heat (q) to raise the temperature of 100.0 g (m) of the metal from 25.0 °C (T_i) to 55.0 °C (T_f)?

$$q = m \cdot c \cdot \Delta T$$

$$1155\ \text{J} = (100.0\ \text{g}) \cdot c \cdot (55.0\,°\text{C} - 25.0\,°\text{C}) \quad ; \quad \Delta T = T_f - T_i$$

$$c = \frac{1155\ \text{J}}{(100.0\ \text{g}) \cdot (30.0\,°\text{C})}$$

$$c = 0.385\ \frac{\text{J}}{\text{g} \cdot °\text{C}}$$

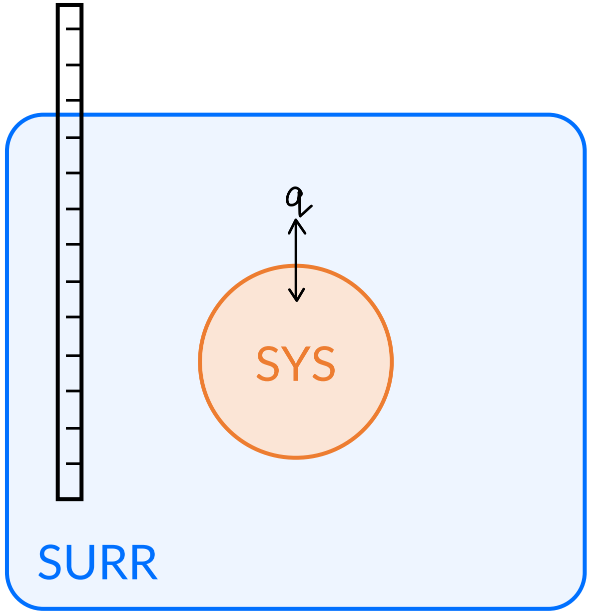
What is the identity of the metal? from table on previous slide: copper

Calorimetry: Measuring Energy Changes

6.6

Calorimetry: The study of heat transfer between substances by measuring the temperature changes of the substances involved.

- The heat released or absorbed by the system is determined by measuring the temperature change of the surroundings.



Start from first law (conservation of energy):

$$q_{sys} + q_{surr} = 0 \quad \text{(zero)}$$

$$q_{sys} = -q_{surr}$$

MISCONCEPTION: This negative sign arises algebraically from the first law. It has nothing to do, necessarily, with exothermicity.

In other words, measure q_{surr} to understand q_{sys}!

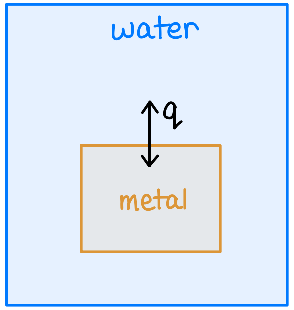
IN-CLASS QUESTION 1

Learning Objective(s): 6.7

A $\overset{m_{metal}}{\text{25.0 g block of metal}}$ initially at $\overset{T_{i,metal}}{\text{95.0 °C}}$ is placed in a calorimeter containing $\overset{m_{water}}{\text{121 g of water}}$ initially at $\overset{T_{i,water}}{\text{27.2 °C}}$. Assuming that no heat is lost to the calorimeter or the surroundings, what is the final temperature (in °C) of the water and metal?

$$c_{\text{metal}} = 0.49\ \frac{\text{J}}{\text{g} \cdot {}^\circ\text{C}}$$

$$c_{\text{water}} = 4.184\ \frac{\text{J}}{\text{g} \cdot {}^\circ\text{C}}$$



$m_{metal} = 25.0\ g$ $\quad$ $m_{water} = 121\ g$

$c_{metal} = 0.49\ \frac{J}{g \cdot °C}$ $\quad$ $c_{water} = 4.184\ \frac{J}{g \cdot °C}$

$T_{i,metal} = 95.0\,°C$ $\quad$ $T_{i,water} = 27.2\ °C$

$T_f = ?$ $\quad$ $T_f = ?$

Start from first law (conservation of energy)

$$q_{metal} + q_{water} = 0 \quad \text{(zero)}$$

$$q_{metal} = -q_{water}$$; Replace each side with $q = m \cdot c \cdot \Delta T$

$$m_{metal} \cdot c_{metal} \cdot (T_f - T_{i,metal}) = -\, m_{water} \cdot c_{water} \cdot (T_f - T_{i,water})$$; T_f always same for "objects"

$$(25.0\ g)\left(0.385\ \frac{J}{g \cdot °C}\right) \cdot (T_f - 95.0\,°C) = -(121\ g)\left(4.184\ \frac{J}{g \cdot °C}\right) \cdot (T_f - 27.2\ °C)$$

$$\Big\downarrow \text{algebra}$$

$$T_f = 28.8\,°C$$

Calorimetry: Measuring Energy Changes

6.6

Constant pressure calorimetry takes place on the benchtop where the atmospheric pressure is relatively constant such that $q_{\mathrm{P,rxn}}$ (and ΔH_{rxn}) can be measured.

- The reaction (rxn) is the system and the solution (soln) in which the reaction takes place is the surroundings.

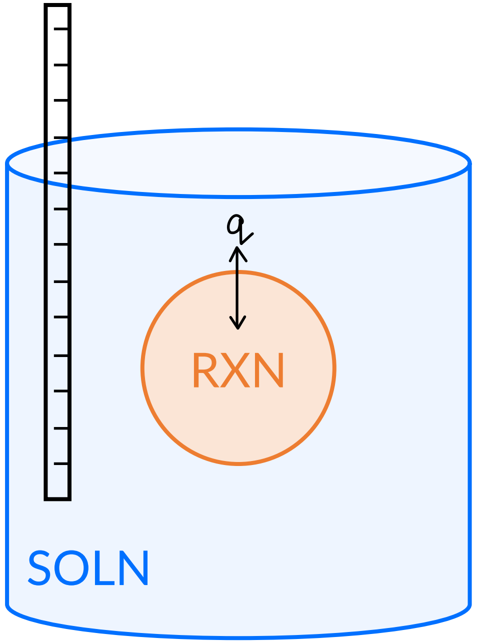
Start from first law (conservation of energy):

$$q_{rxn} + q_{soln} = 0 \quad \text{(zero)}$$

$$q_{rxn} = -q_{soln}$$

MISCONCEPTION: This negative sign arises algebraically from the first law. It has nothing to do, necessarily, with exothermicity.

In other words, measure q_{soln} to understand q_{rxn}!

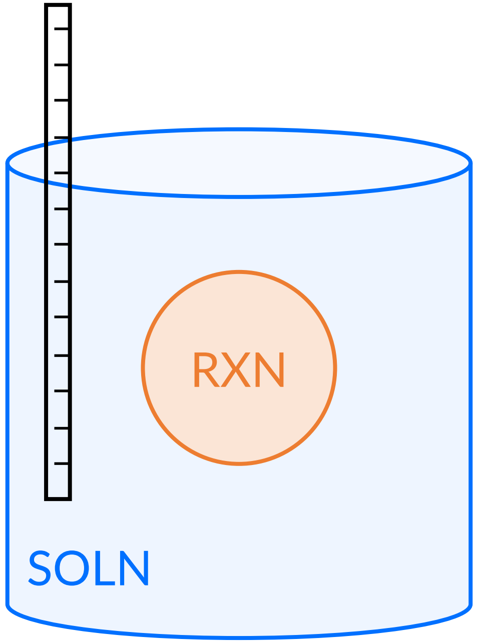


Calorimetry: Measuring Energy Changes

6.6

Remember that "system" and "surroundings" are fictitious designation.

In an *aqueous* reaction, we measure the temperature of the solution and the solution is part of the surroundings.



Temperature of solution	Surroundings (soln)	System (rxn)	ΔH_{rxn}
Increases	absorbs heat	releases heat	$\Delta H_{rxn} < 0$ exothermic
Decreases	releases heat	absorbs heat	$\Delta H_{rxn} > 0$ endothermic

IN-CLASS QUESTION 2

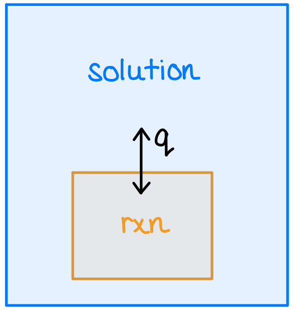
Learning Objective(s): 6.7

A 30.0 mL sample of 0.500 M NaOH (base) and a 10.0 mL sample of 0.500 M HCl (acid) are combined in a constant pressure calorimeter. The temperature of the combined solutions increases from 20.40 °C to 23.55 °C. Assuming the calorimeter loses no heat to the surroundings, what is ΔH_{rxn} in kJ/mol?

$\Delta T_{soln} = (T_f - T_i)_{soln}$

Assume the density of the solution is 1 g/mL and the specific heat capacity is that of water.

Draw a picture!



what is rxn?

Just solutes!

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(aq)$

what is soln?

All the water from the aqueous solutions!

$30.0\text{ mL} + 10.0\text{ mL} = 40.0\text{ mL} \times \frac{1\text{ g}}{1\text{ mL}} = 40.0\text{ g soln}$

Start from first law (conservation of energy)

$q_{rxn} + q_{soln} = 0$ (zero)

$q_{rxn} = -q_{soln}$; $q = m \cdot c \cdot \Delta T$

$= -m_{soln} \cdot c_{soln} \cdot (T_f - T_i)_{soln}$

$= -(40.0\text{ g}) \cdot \left(4.184 \frac{\text{J}}{\text{g} \cdot {}^\circ\text{C}}\right) \cdot (23.55\,^\circ\text{C} - 20.40\,^\circ\text{C})$

$q_{rxn} = -527._2\text{ J}$ total amount released

What is n_{rxn} (mol of rxn)? Just another stoichiometric ratio!

$1\,NaOH + 1\,HCl \rightarrow 1\,NaCl + 1\,H_2O$

1 mol rxn : 1 mol NaOH : 1 mol HCl : 1 mol NaCl : 1 mol H_2O : ΔH_{rxn}

Start with LR! In this case, HCl.

$0.0100\text{ L HCl} \times \frac{0.500\text{ mol HCl}}{1\text{ L}} \times \frac{1\text{ mol rxn}}{1\text{ mol HCl}} = 0.00500\text{ mol rxn}$

Scale q_{rxn} by n_{rxn}: $\Delta H_{rxn} = \frac{q_{rxn}}{n_{rxn}} = \frac{-0.527_2\text{ kJ}}{0.00500\text{ mol rxn}} = -105 \frac{\text{kJ}}{\text{mol}}$ (exo)

Enthalpy in Chemical Reactions

6.7

Thermochemical equation

- Coupling a balanced chemical equation with its corresponding ***change in enthalpy***, $\boldsymbol{\Delta H_{\mathrm{rxn}}}$ (also called ***heat of reaction*** or ***enthalpy of reaction***).
- Thermochemical equations are *only* interpreted as mole-mole ratios.
- For example, we'd say that:

"For every one mole of reaction, one mole of $C_6H_{12}O_6$ reacts with six moles of O_2 to produce six moles of CO_2 and six moles of H_2O, and releases 3500 kJ of heat."

$$C_6H_{12}O_6(\ell) + 6\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(\ell) \qquad \Delta H_{\mathrm{rxn}} = -3500\ \frac{\mathrm{kJ}}{\mathrm{mol}}$$

exothermic; treat heat like "product"

1 mol rxn : 1 mol $C_6H_{12}O_6$: 6 mol O_2 : 6 mol CO_2 : 6 mol H_2O : −3500 kJ

treat as conversion factors

$$\frac{-3500\ \mathrm{kJ}}{1\ \mathrm{mol\ rxn}} \text{ or } \frac{-3500\ \mathrm{kJ}}{1\ \mathrm{mol}\ C_6H_{12}O_6} \text{ or } \frac{-3500\ \mathrm{kJ}}{6\ \mathrm{mol}\ O_2} \text{ or } \frac{-3500\ \mathrm{kJ}}{6\ \mathrm{mol}\ CO_2} \text{ or } \frac{-3500\ \mathrm{kJ}}{6\ \mathrm{mol}\ H_2O}$$

Enthalpy in Chemical Reactions

6.7

Thermochemical equations are also *only* interpreted to represent the enthalpy change for the reaction ***exactly* as it is written from left to right**.

$$C_6H_{12}O_6(\ell) + 6\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(\ell) \qquad \Delta H_{rxn} = -3500\,\frac{\text{kJ}}{\text{mol}}$$

- The units for ΔH_{rxn} are kJ/mol or J/mol and interpreted as the quantity of heat per mole of reaction (rxn).
- ΔH_{rxn} is an *intensive property* specific to a reaction, so it is a constant.
- If you carry out more or less than one mole of reaction, then the quantity of heat (q_{rxn}) associated with the experiment carried out changes/scales.

$$\Delta H_{rxn} = \frac{q_{rxn}}{n_{rxn}} \quad \text{or} \quad q_{rxn} = n_{rxn} \times \Delta H_{rxn}$$

IN-CLASS QUESTION 3

Learning Objective(s): 6.8

The overall reaction in a commercial heat pack can be represented as:

$$4\,Fe(s) + 3\,O_2(g) \rightarrow 2\,Fe_2O_3(s) \qquad \Delta H^o_{rxn} = -1652\,\frac{kJ}{mol}$$ exothermic

What is the total heat (q_{rxn} in kJ) associated with the reaction between 2.00 g of Fe(s) and *excess* $O_2(g)$?

This is just stoichiometry with heat!

$\frac{-1652\text{ kJ}}{1\text{ mol rxn}}$ or $\frac{-1652\text{ kJ}}{4\text{ mol Fe}}$ or $\frac{-1652\text{ kJ}}{3\text{ mol }O_2}$ or $\frac{-1652\text{ kJ}}{2\text{ mol }Fe_2O_3}$; $\Delta H_{rxn} = \frac{q_{rxn}}{n_{rxn}}$ or $q_{rxn} = n_{rxn} \times \Delta H_{rxn}$

Method 1: Use n_{rxn}

$$2.00\text{ g Fe} \times \frac{1\text{ mol Fe}}{55.845\text{ g Fe}} \times \frac{1\text{ mol rxn}}{4\text{ mol Fe}} \times \frac{-1652\text{ kJ}}{1\text{ mol rxn}} = -14.8\text{ kJ}$$

Method 2: Use Fe directly

$$2.00\text{ g Fe} \times \frac{1\text{ mol Fe}}{55.845\text{ g Fe}} \times \frac{-1652\text{ kJ}}{4\text{ mol Fe}} = -14.8\text{ kJ}$$

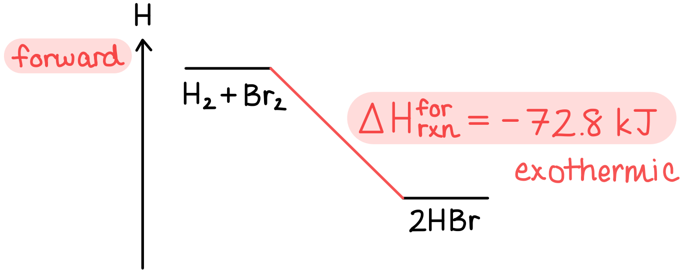
EXAMPLE PROBLEM

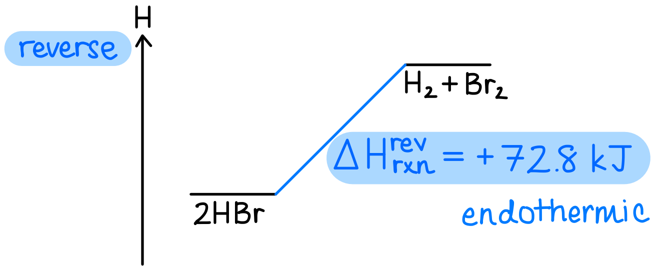
Learning Objective(s): 6.8

reverse rxn!

How much energy, as heat in kJ, would be needed to *decompose* 9.74 g of HBr(g) into its elements?

$$H_2(g) + Br_2(g) \rightarrow 2\ HBr(g) \qquad \Delta H^{o}_{rxn} = -72.8\ \frac{kJ}{mol}$$





Method 1: Use n_{rxn}

$$9.74\ g\ HBr \times \frac{1\ mol\ HBr}{80.912\ g\ HBr} \times \frac{1\ mol\ rxn}{2\ mol\ HBr} \times \frac{+72.8\ kJ}{1\ mol\ rxn} = 4.38\ kJ$$

Method 2: Use HBr directly

$$9.74\ g\ HBr \times \frac{1\ mol\ HBr}{80.912\ g\ HBr} \times \frac{+72.8\ kJ}{2\ mol\ HBr} = 4.38\ kJ$$

BONUS QUESTION

Assessed on: *undetermined*

The overall reaction in a commercial heat pack can be represented as:

$$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s) \qquad \Delta H^{o}_{\mathrm{rxn}} = -1652\,\frac{\mathrm{kJ}}{\mathrm{mol}}$$

What is the total heat (q_{rxn} in kJ) associated with the reaction between 515 g of Fe(*s*) and 475 g O_2(*g*)?